



Argon ONE V5 Case with M.2 NVMe SSD Slot

Product Datasheet

Product Name	Argon ONE V5 Case with M.2 NVMe SSD Slot
Product Model	A24-005-AR-V5-NVME-S
EAN / UPC	5065019946164
Data Sheet Version	Rev 8 (build v8)
Date	18 August 2026
Argon 40 SKU	ARG-ONE-V5-NVME

Overview

Aluminum Raspberry Pi 5 case with M.2 NVMe SSD slot via PCIe for maximum storage performance.

1. Application

- Raspberry Pi 5

Product Features

- M.2 NVMe SSD slot (PCIe, 2230/2242/2280)
- Full-size HDMI ports (2x)
- PWM-controlled fan
- Programmable power button
- 40-pin GPIO passthrough header
- Front 2x USB A Ports
- Front 3.5mm Audio IN/OUT
- 30mm blower-type PWM fan
- U-shaped PCIe connector for M.2 board
- Aluminum THRMML heatsinks with blue thermal pads included
- 2x USB 3.0 ports (passthrough)
- Gigabit Ethernet (passthrough)
- Programmable safe shutdown via power button
- Magnetic GPIO header cover
- Injected ABS plastic case bottom
- M.2 form factor support: 2230, 2242, 2260, 2280
- U-shaped PCIe connector with locking tabs
- Silicon thermal pads for CPU, RP1, PMIC, RAM, and NVMe
- FE1.1S USB Hub Chip for front USB 2.0 ports
- Black NVMe supplemental power cable included
- Comprehensive NVMe troubleshooting documentation

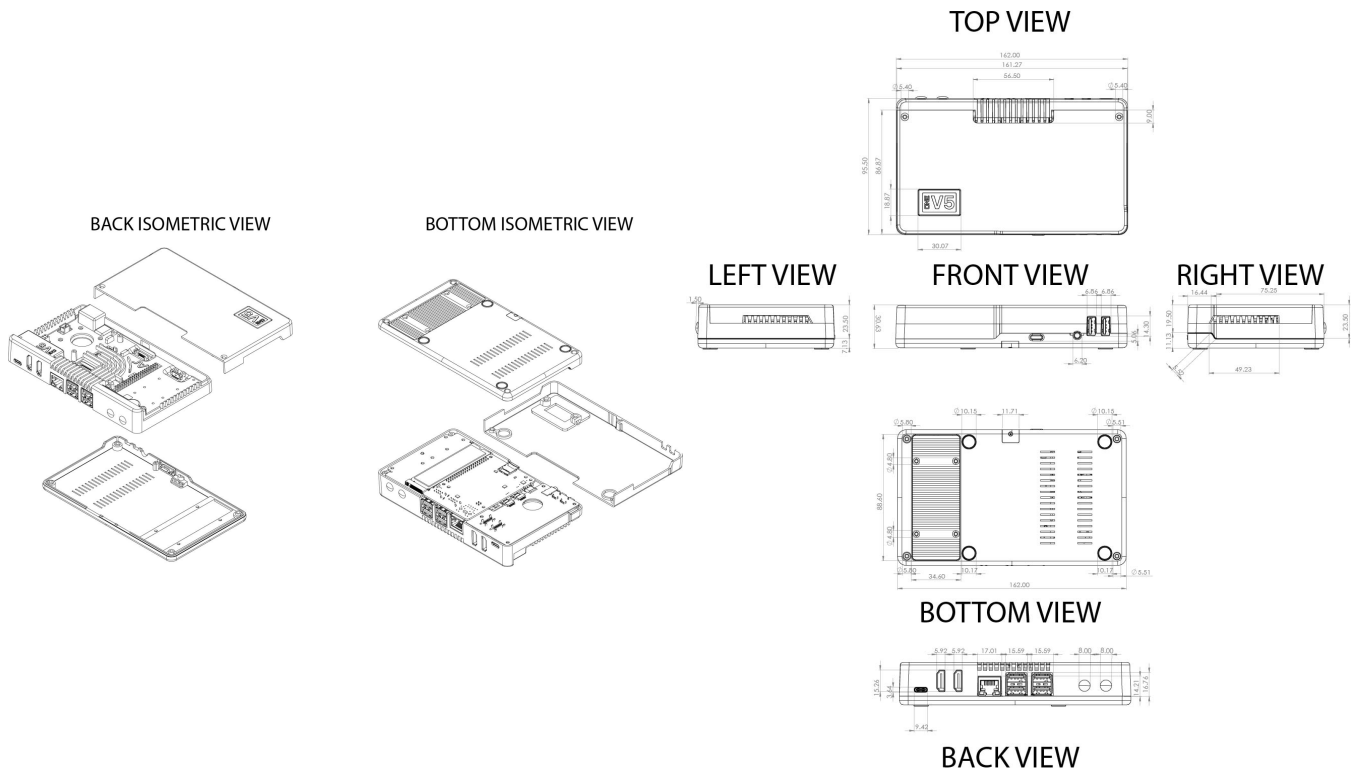
2. Product Specifications

- **Category:** ONE v5
- **Weight (kg):** 0.355
- **Dimensions (cm):** 10.00 x 4.00 x 16.50 cm
- **HTS code:** 8473.30.5100
- **active_cooling:** 30mm blower-type PWM fan, software-controlled

- **audio:** Built in DAC with MIC Powered by HS100B - DAC Chip
- **audio_connector:** 3.5mm audio out and mic in
- **audio_dac:** HS100B - DAC Chip
- **audio_io:** 3.5mm audio out + 3.5mm mic in via HS100B DAC Chip
- **audio_output:** 3.5mm audio out and mic in
- **bootloader_minimum:** June 2026 or newer
- **bootloader_minimum_date:** June 2026
- **bootloader_minimum_version:** June 2026 or newer
- **bottom_shell_material:** Injected ABS plastic
- **cable_type:** PCIe ribbon (FPC) cable
- **case_material:** Injected aluminum alloy top and injected ABS plastic bottom
- **case_material_bottom:** Injected ABS plastic
- **case_material_top:** Injected aluminum alloy
- **ce_certified:** Yes
- **compatible_case:** Argon ONE V5
- **compatible_products:** Argon ONE V5, Argon ONE V3 (NVMe), Argon NEO 5, Argon Industria PoE+ HAT
- **compatible_raspberry_pi:** Raspberry Pi 5
- **compatible_sbc:** Raspberry Pi 5
- **connector_type:** U-shaped PCIe connector for M.2 board
- **cooling_fan:** Blower type 30mm PWM actively controlled by the Raspberry Pi OS
- **cooling_type:** Blower type 30mm PWM actively controlled by the Raspberry Pi OS
- **dac_chip:** HS100B DAC Chip
- **dimensions:** 110mm x 90mm
- **dimensions_cm:** 10.00 × 4.00 × 16.50
- **dual_gen2_chipset:** ASM1182e
- **dual_gen3_chipset:** ASM2806
- **ethernet:** Ethernet Port
- **fan:** PWM 30mm Fan
- **fan_type:** 30mm blower-type PWM, software-controlled
- **flat_head_screws:** 8 pcs Flat head 2.5x6 screws
- **form_factor:** Designed exclusively for Raspberry Pi 5
- **front_usb_2_0_ports:** 2× USB 2.0 via FE1.1S USB Hub Chip
- **front_usb_ports:** 2x USB 2.0 Ports
- **gen_2_pcie_switch_chip:** ASM1182e
- **gen_3_pcie_switch_chip:** ASM2806
- **gigabit_ethernet:** Passthrough from Raspberry Pi 5
- **gpio:** Full 40-pin passthrough header with magnetic cover
- **gpio_access:** Full GPIO Access with Magnetic cover
- **gpio_header:** 40-pin passthrough with magnetic GPIO header cover
- **gpio_pins:** first 8 GPIO pins
- **hdmi:** 2x full-size HDMI
- **hdmi_ports:** 2 Regular HDMI
- **hts_code:** 8473.30.5100
- **interface_type:** PCIe NVMe
- **m2_compatible_sizes:** 2230, 2242, 2260, 2280
- **m2_form_factors:** 2230, 2242, 2260, 2280
- **m2_slot:** M.2 NVMe PCIe, 2230/2242/2280
- **m2_slot_type:** PCIe 2.0 ×1 (U-shaped connector)
- **material:** Aluminum alloy
- **minimum_bootloader_version:** June 2026 or newer
- **minimum_psu:** 5V / 3A (15W)
- **model_number:** A24-005-AR-V5-NVME-S
- **nvme_board_variants:** Single (Gen 2), Dual Gen 2, Quad Gen 2, Dual Gen 3, Quad Gen 3

- **oled_dimensions:** 55mmx90mm
- **oled_module_display:** 1.8 inch
- **optional_oled:** 1.8 inch OLED Module
- **optional_zigbee:** CC2562P Zigbee Chip for Home Assistant
- **passive_cooling:** Aluminum alloy case body + aluminum THRML heatsinks with blue silicon thermal pads
- **pcie_connector_type:** U-shaped with locking tabs
- **pcie_interface:** PCIe 2.0 ×1
- **platform:** Raspberry Pi 5
- **power_button:** Programmable; supports safe shutdown via software
- **power_input:** USB-C
- **product_dimensions:** 10.00 × 4.00 × 16.50 cm
- **product_name:** Argon Industrial V5 ZIGBEE Module
- **product_weight:** 0.355 kg
- **quad_gen3_chipset:** ASM2806 PCIe Gen3 Packet Switching Chips
- **recommended_os:** Raspberry Pi OS Trixie (Desktop)
- **recommended_psu:** 5V / 5A (27W Power Delivery)
- **required_os:** Raspberry Pi OS Trixie
- **rohs_compliant:** Yes
- **round_head_screws:** 4 pcs Round head 2.5x8 screws
- **single_drive_variant:** RPI 5 Native 1 PCIe Gen2
- **sku:** ARG-ONE-V5-NVME
- **ssd_form_factor:** M.2 NVMe
- **storage_expansion:** M.2 NVMe-PCIe Expansion Boards with THRML Cooling
- **supported_m2_form_factors:** 2230, 2242, 2280
- **thermal_pad_types:** Transparent (CPU/RP1/PMIC/RAM), Blue (NVMe)
- **thermal_pads_included:** True
- **top_shell_material:** Injected aluminum alloy
- **typical_pcie_speed:** 400–500 MB/s
- **upc_ean:** 5065019946164
- **usb_2_0_front:** 2× USB 2.0 ports via FE1.1S USB Hub Chip
- **usb_2_0_ports:** 2× (via FE1.1S USB Hub Chip)
- **usb_2_ports:** 2x USB 2.0 Ports
- **usb_3_0_passthrough:** 2× USB 3.0 (from Raspberry Pi 5)
- **usb_3_0_ports:** 2× USB 3.0 passed through from Raspberry Pi 5
- **usb_3_ports:** 2x USB 3.0 Ports
- **usb_hub_chip:** FE1.1S - USB Hub Chip
- **weight:** 0.355 kg
- **weight_kg:** 0.355
- **zigbee_chip:** CC2562P

3. Mechanical Information and Functional Diagram



Illustration(s) reused from the existing marketing datasheet on file.

4. Reference Design

- (Electronics Schematics)

5. Product Compliance Information

- This part is compliant with EU 2002/95/EC RoHS directive (Restrictions on the Use of Certain Hazardous Substances in Electrical and Electronic Equipment)

6. Contact Information

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Document Control

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